PROJECT - 01

PhonePe Pulse

PhonePe Transaction Insights

PROJECT DOCUMENTATION

SUBMITTED BY: Sruthi G
BATCH: AIML-C-WD-E-B48

Problem Statement:

With the increasing reliance on digital payment systems like PhonePe, understanding the dynamics of transactions, user engagement, and insurance-related data is crucial for improving services and targeting users effectively. This project aims to analyze and visualize aggregated values of payment categories, create maps for total values at state and district levels, and identify top-performing states, districts, and pin codes.

Problem solving:

1. **Data Sourcing -** Cloned data to local from GitHub.

2. **Extract -** Each JSON file was read using Python's json module. The folder structure (state/year/quarter) was traversed using os.listdir() to extract state, year, and quarter metadata directly from the file path.

3. **Transfer -** Organized all extracted data into Pandas DataFrames with consistent column naming.

4. **Load -** A reusable helper function insert_df() was built to insert any DataFrame into its corresponding MySQL table using parameterized queries with %s placeholders — preventing SQL injection. All 9 tables were loaded successfully using this single function.

Business Case Studies

Five business case studies were selected and analyzed from the PhonePe transaction data. Each study includes a scenario, SQL query approach, visualization type, and key insights.

Case Study 1 — Decoding Transaction Dynamics

- Scenario: Identify variations in transaction behavior across states, quarters, and payment categories
- Analysis: Top 10 states by transaction amount, payment type breakdown, year-wise growth trend
- Visualization: Horizontal bar chart, pie chart, line chart
- **Insight:**

  --> Maharashtra, Telangana, and Karnataka consistently lead in transaction volumes.

  --> There is a decline in the transaction volumes of andaman & nicobar islands, ladakh and lakshadweep.

  --> Merchant payments dominate at over 55.4% of all transactions.

  --> As for the years, the transaction growth is accelerating over time.

Case Study 2 — Device Dominance & User Engagement

- Scenario: Understand user preferences across different device brands by region
- Analysis: Top mobile brands by user count, app opens by state
- Visualization: Vertical bar chart, horizontal bar chart
- **Insight:**

--> Xiaomi and Samsung dominate across most states.

--> Lava, HMD Global, Infinix show significant downfall.

--> Karnataka, Maharashtra, Andhra Pradesh show significantly higher app engagement relative to registration numbers.

--> Andaman & Nicobar islands, mizoram, meghalaya show significantly lower app engagement relative to registration numbers.

Case Study 3 — Insurance Penetration & Growth

- Scenario: Analyze insurance growth trajectory and identify untapped opportunities at state level
- Analysis: Insurance growth by year, top states by insurance policies
- Visualization: Line chart, horizontal bar chart
- **Insight:**

--> Insurance adoption shows consistent year-on-year growth.

--> Several northeastern states-Andaman & nicobar islands, Sikkim and Meghalaya show very low adoption indicating major untapped market potential.

--> Insurance adoptation is more in Maharashtra, Andhra pradesh

Case Study 4 — Transaction Analysis for Market Expansion

- Scenario: Identify top and bottom performing districts for strategic market expansion
- Analysis: Top 10 districts by transaction volume, bottom 10 districts as expansion opportunities
- Visualization: Horizontal bar charts
- **Insight:**

--> Bengaluru Urban, Pune, Central dominate across most districts.

--> Dibang valley, kmale, Pherzawl show significant downfall.

Case Study 5 — User Engagement & Growth Strategy

- Scenario: Analyze user engagement across states to develop growth strategies
- Analysis: Registered users vs app opens scatter analysis, top states by registered users
- Visualization: Scatter plot, vertical bar chart
- **Insight:**

--> Maharashtra, Uttar pradesh states have high registrations and high app opens .

--> Lakshadweep, Mizoram states have high registrations but low app opens.

--> Delhi, Karnataka, uttar pradesh tops by maximum number of registered users.

--> Lakshadweep, Mizoram, Andaman & nicobar islands have minimum number of registered users.

StreamLit Dashboard:-

To visualize a large amount of data , I used the StreamLit dashboard. It acts as a bridge for the data.

Charts- Horizontal bar,
Vertical bar,
Line chart,
Pie chart,
Scatter plot

For extracting the contents of the chart for the given case studies tables such as

aggregated_transactions for transaction dynamics,
aggregated_users and map_users for device and user engagement,
aggregated_insurance and map_insurance for insurance penetration,
map_transactions for market expansions,
map_users and top_users for user growth strategy.

Choropleath mapping was also done in the streamlit dashborad.
Run command for StramLit- **streamlit run dashboard.py**

Choropleath Mapping:-

--> I have created a choropleath Indian map using the **geoJson** for drawing the map shape, **MySql** for state name column to match against geoJSON state names for colouring the map.

--> Three maps were built

1. State VS Total Transactions
2. State VS Total Users
3. State VS Total Insurance

--> The darker colour on the map shows the higher value column .

--> **Insight**

1. It identifies economic hotspots(**map 1**)
2. It shows digital adoption geography(**map 2**)
3. It reveals penetration gaps (**map 3**)

Key Insights and findings for the 5 case studies:-

Case 1-

Reason - As we can see from the charts, the bottom 10 are mostly northeastern states. They are the rural states that are still under development. Poor internet connections, using less smartphones, less awareness might be some of the reasons for the downfall.

Solution - Running awareness programs, giving cashback offers for the first time payments might help them.

Case 2-

Reason - Some people might install phonepe for single transaction and never open again. This might be due to poor UIs in the phone brands. Too many notifications might make them never open the app again. Low ram in the phones might make it difficult for them to use the application.

Solution - Building phonepe-lite app for brands with low ram, running campaigns that targets the dormant users.

Case 3-

Reason - Same as the case 1, the northeastern states are still under development, they might not have the awareness, they might not trust the digital payments unlike urban cities like bangalore, maharashtra and more.

Solution - Creating awareness programs and campaigns for the people. Conduction camps to help them understand and use the digital payments methods and having regional language support.

Case 4 -

Reason - Having less/no internet connection might make it difficult for people to use the digital payment method. Some rural towns might not still implement QR payments due to less trust. people being used to do direct cash transactions.

Solution - Providing telecom data(4G) at offer prices, providing live reviews using advertisement might make them to trust digital transactions.

Case 5 -

Reason - Average smartphones might make it difficult to do transactions. People might prefer government owned digital transaction apps more than phonepy. In areas with weak telecom signals, the SMS OTP for PhonePe login or UPI PIN reset frequently fails to arrive on time .

Solution - Earning cashbacks, Using missed call verification instead of SMS OTP so users in weak network areas can log in without getting stuck, providing offers for maintaing streak.

Skills Acquired:-

1. Data extraction from nested JSON structures and GitHub repositories
2. ETL pipeline design using Python and MySQL
3. SQL proficiency — table design, data types, INSERT, SELECT, GROUP BY, ORDER BY
4. Data cleaning and transformation using Pandas.
5. Interactive data visualization.
6. Dashboard development using Streamlit.
7. Version control using Git and GitHub
8. Analytical thinking and business insight generation